Skills Summary

Reversed and Non-Equivalent Ratios

Use this reference while completing your worksheet or when studying.

Skill 1 — The order is part of the answer

Rule: "the ratio of A to B" means $\frac{A}{B}$ — the quantity named FIRST goes on top, whatever order the numbers appeared in the sentence.

Sense check: if the top quantity is the bigger one, the ratio must come out greater than 1. A ratio below 1 when you expected "more" is a reversed ratio.

Example: A shelter has 10 dogs and 15 cats. Write the ratio of cats to dogs.
Step 1. Cats are named first, so cats go on top even though dogs were counted first in the sentence.
Step 2. $\frac{15}{10}$, and dividing both parts by 5 gives $\mathbf{\frac{3}{2}}$ — greater than 1, as it should be with more cats.

Try it: A box holds 9 red pens and 12 blue pens. Write the ratio of blue to red in simplest form.

check: $\mathbf{\frac{4}{3}}$

Skill 2 — Scale both parts by the same number

Rule: to build an equivalent ratio, multiply (or divide) *both* parts by the same number. Find the factor from the part you know: factor = new value ÷ old value.

Adding does not work. 2 : 3 and 4 : 5 are not equivalent, even though each part grew by 2.

Example: A table starts at 3 : 5. The first part becomes 12. What is the second part?
Step 1. Find the factor from the part that changed, then apply that same factor to the other part.
Step 2. $12 \div 3 = 4$, so the second part is $5 \times 4 = \mathbf{20}$.

Try it: A table starts at 4 : 7. The first part becomes 20. What is the second part?

check: **35**

Skill 3 — Testing two ratios for equivalence

Three tests, all equivalent: simplify both and compare; compare the unit rates (divide top by bottom); or cross multiply and see whether the two products match. If any test says no, the ratios are not equivalent.

Example: Are $6 : 9$ and $10 : 15$ equivalent?
Step 1. Both parts of each ratio share a factor, so simplifying is the quickest of the three tests here.
Step 2. $\frac{6}{9} = \frac{2}{3}$ and $\frac{10}{15} = \frac{2}{3}$, so $=$ — they are equivalent.

Try it: Write $<$, $=$, or $>$ between $\frac{8}{12}$ and $\frac{10}{16}$.

check: $>$